

Mathematics for Machine Learning

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Theorem

Let $a < c < b$. If $f \in \mathcal{R}[a, c]$ and $f \in \mathcal{R}[c, b]$, then $f \in \mathcal{R}[a, b]$.

Definite Integral

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$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

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Theorem

If f is monotone on $[a, b]$ then $f \in \mathcal{R}[a, b]$.

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If $f, g \in \mathcal{R}[a, b]$, then

① $f \pm g \in \mathcal{R}[a, b]$ and

$$\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx,$$

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③ $|f|, fg \in \mathcal{R}[a, b]$.

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If $f, g \in \mathcal{R}[a, b]$ and $f(x) \geq g(x)$ for all $x \in [a, b]$, then

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If $f \in \mathcal{R}[a, b]$ then

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

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If f is continuous at the point $x_0 \in [a, b]$, then F is differentiable and

$$F'(x_0) = f(x_0).$$

Theorem

If $f \in C[a, b]$ and F is an antiderivative of f , then

$$\int_a^b f(t) dt = F(b) - F(a)$$

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$$\int_a^b f(t) dt = F(b) - F(a) = F(x) \Big|_{x=a}^b.$$

Theorem

Let $f \in C[a, b]$, $\varphi : [\alpha, \beta] \rightarrow [a, b]$, $\varphi \in C^1[\alpha, \beta]$ and $\varphi(\alpha) = a$, $\varphi(\beta) = b$, then

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt$$

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Theorem

If $f, g \in C^1[a, b]$, then

$$\int_a^b f dg = fg|_a^b - \int_a^b g df$$

Improper Integral

Definition

Let $f : [a, +\infty) \rightarrow \mathbb{R}$ and $f \in \mathcal{R}[a, A]$ for all $A > a$. If there exists the limit

$$\lim_{A \rightarrow +\infty} \int_a^A f(x) dx,$$

then this limit is called the improper integral of f from a to $+\infty$ and it is denoted by

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In the same way we can define $\int_{-\infty}^a f(x) dx$.

Improper Integral

Example

$$\int_0^{+\infty} \frac{dx}{1+x^2} =$$

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$$\int_0^{+\infty} \frac{dx}{1+x^2} = \lim_{A \rightarrow +\infty} \operatorname{arctg} x \Big|_0^A = \lim_{A \rightarrow +\infty} \operatorname{arctg} A = \frac{\pi}{2}.$$

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Proposition

If $\int_a^{+\infty} f(x) dx$ is convergent, then $\lim_{A \rightarrow \infty} \int_A^{+\infty} f(x) dx = 0$.

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Example

For the function

$$f(x) = \begin{cases} 1, & x \in \mathbb{N}, \\ 0, & x \notin \mathbb{N}, \end{cases}$$

the integral $\int_0^{+\infty} f(x) dx$ is convergent, but $\lim_{x \rightarrow +\infty} f(x) \neq 0$.

Improper Integral

Theorem

If $f(x) \geq 0$, then the integral $\int_a^{+\infty} f(x) dx$ is convergent if and only if the function $F(A) = \int_a^A f(x) dx$ is bounded above. If it is not bounded above then $\int_a^{+\infty} f(x) dx = +\infty$.

Improper Integral

Theorem

Let $f, g : [a, +\infty) \rightarrow \mathbb{R}$ and $f, g \geq 0$. If there exists a number $A > a$ such that $f(x) \leq g(x)$ for all $x > A$, then from the convergence of the integral

$\int_a^{+\infty} g(x) dx$ follows that $\int_a^{+\infty} f(x) dx$ is convergent too.

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Theorem

Let $f, g : [a, +\infty) \rightarrow \mathbb{R}$ and $f, g \geq 0$. If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = K$, where

$0 \leq K < +\infty$, then from the convergence of the integral $\int_a^{+\infty} g(x) dx$

follows that $\int_a^{+\infty} f(x) dx$ is convergent too.

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$$\text{a) } \int_0^1 \frac{1}{\sqrt{x}} dx =$$

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a) $\int_0^1 \frac{1}{\sqrt{x}} dx = 2\sqrt{x} \Big|_0^1 = 2.$

b) *Prove that the gamma function $\Gamma(\alpha)$, defined as*

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

satisfies the relation

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1), \alpha > 1.$$

Taylor Series

Definition

The **Taylor polynomial** of degree n of $f : \mathbb{R} \rightarrow \mathbb{R}$ at x_0 is defined as

$$T_n(x) := \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

where $f^{(k)}(x_0)$ is the k th derivative of f at x_0 (which we assume exists) and $\frac{f^{(k)}(x_0)}{k!}$ are the coefficients of the polynomial.

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Definition

For a smooth function $f \in C^\infty$, $f : \mathbb{R} \rightarrow \mathbb{R}$ the **Taylor series** of f at x_0 is defined as

$$T_\infty(x) := \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

Remark

In general, a Taylor polynomial of degree n is an approximation of a function, which does not need to be a polynomial. The Taylor polynomial is similar to f in a neighborhood around x_0 .

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Example

Find the Taylor polynomial $T_5(x)$ of $f(x) = x^3$ at point $x_0 = 2$. Verify that $T_5(x) = f(x)$.

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Example

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Remark

Taylor polynomial of degree n is an exact representation of a polynomial f of degree $k \leq n$.

Taylor Series

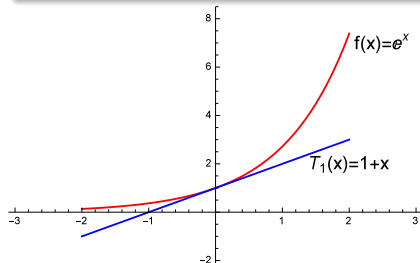
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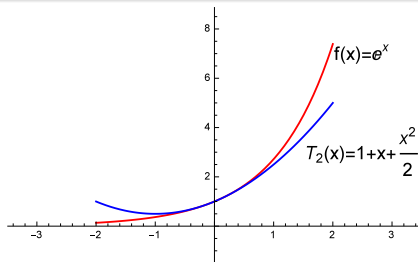
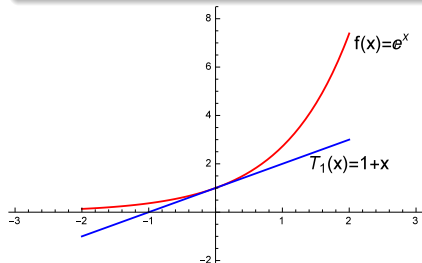
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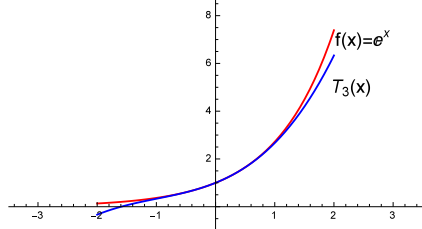
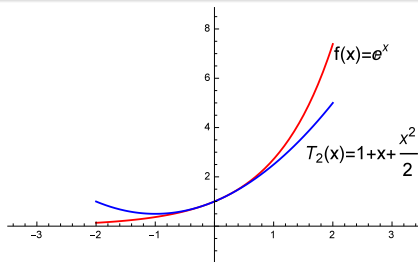
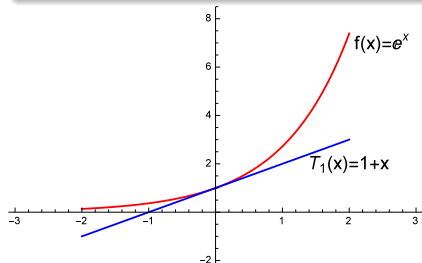
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Taylor's theorem

If $|f^{(n+1)}(x)| \leq M$ for $x \in (x_0 - h, x_0 + h)$, then

$$f(x) = T_n(x) + r_n(x), \text{ where } |r_n(x)| \leq \frac{M}{n!} |x - x_0|^{n+1},$$

for $x \in (x_0 - h, x_0 + h)$.

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$$\textcircled{5} \quad (1+x)^\alpha = \sum_{n=0}^{\infty} \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n, \quad x \in (-1, 1).$$

Partial Differentiation and Gradients

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$$\begin{aligned} f'_{x_1}(\mathbf{x}) &= \frac{\partial f}{\partial x_1}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h} \\ &\vdots \\ f'_{x_n}(\mathbf{x}) &= \frac{\partial f}{\partial x_n}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(x_1, x_2, \dots, x_n + h) - f(x_1, x_2, \dots, x_n)}{h}. \end{aligned}$$

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The row vector

$$\nabla f = \frac{df}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} & \frac{\partial f(\mathbf{x})}{\partial x_2} & \dots & \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{1 \times n},$$

is called the **gradient** of f or the *Jacobian*.